

SCX Self-Evolution: Edge Cases and Failure Modes

Version: 2026-06-28 | **Status:** Formal analysis | **Prerequisite:** Documents 01, 02, 06, 10, 11 **Purpose:** Characterize failure modes of the SCX self-evolution system, deriving conditions under which convergence fails, degrades, or produces pathological behavior. Four canonical failure modes are analyzed in depth, with formal conditions, rates of degradation, and mitigation strategies. **Notation note:** $\mathcal{S}$ denotes the state space (as in core Theorems 1–3). S_t denotes the gatekeeper scoring function at time t (as in Spring self-evolution documents). Ψ denotes the Lyapunov function (renamed from Φ to avoid collision with feature space Φ in Theorem 2). β_t denotes the gatekeeper update rate; η is reserved for the global noise rate.

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1. Failure Mode 1: Premature Convergence (Gatekeeper Freezing)

1.1 Phenomenon

The gatekeeper converges to a suboptimal fixed point S_{sub}^* before the student has learned enough to provide useful feedback. Once frozen, the system cannot escape because the gatekeeper rejects new data that would contradict its (incorrect) judgments.

1.2 Formal Mechanism

The learning rate decays too fast. When β_t (the gatekeeper update rate) decays faster than the student can learn, gatekeeper plasticity is lost before the student provides corrective signal.

Notation note (2026-06-28, B4): The symbol η is reserved throughout the SCX theory for the global noise rate $\eta = P(Z = 1)$. The gatekeeper update rate is denoted β_t . This document previously used $\eta(t)$ locally for the gatekeeper update rate, which has been corrected to β_t for consistency.

Definition 1.1 (Freezing Time). The **freezing time** t_{freeze} is the earliest time such that for all $t \geq t_{\text{freeze}}$:

$$\|\Delta S_t\|_{M_0} \leq \varepsilon_{\text{mach}},$$

i.e., subsequent gatekeeper updates are numerically indistinguishable from zero.

Theorem 12.1 (Freezing Condition — PROVEN). Under condition C6' (two-timescale, $\beta_t = o(\alpha_t)$), if:

$$\beta_t \cdot B_S \leq \varepsilon_{\text{mach}} \quad \text{for } t \geq t_{\text{freeze}},$$

then the gatekeeper freezes. With $\beta_t = \beta_0 t^{-b}$, this occurs at:

$$t_{\text{freeze}} \leq \left(\frac{\beta_0 B_S}{\varepsilon_{\text{mach}}} \right)^{1/b}.$$

Proof. By C7, $\|\Delta S_t\|_\infty \leq \beta_t B_S$. When this bound falls below machine precision, the update is numerically zero. Solving $\beta_0 t^{-b} B_S = \varepsilon_{\text{mach}}$ gives the freezing time. $\square$

Status: PROVEN. This is a direct consequence of C7 and the numerical precision constraint.

1.3 When Freezing is Premature

Freezing is **premature** if $t_{\text{freeze}} < t_{\text{converge}}$, where t_{converge} is the time needed for the student to reach its asymptotic accuracy.

Theorem 12.2 (Premature Freezing Condition — PROVEN). Under strong convexity (SC), the student needs:

$$t_{\text{converge}} = \left(\frac{C_1}{\varepsilon_{\text{target}}} \right)^{1/a}$$

steps to achieve $\mathbb{E}[\|\theta_t - \theta^*\|^2] \leq \varepsilon_{\text{target}}$. Premature freezing occurs when:

$$\boxed{\left(\frac{\beta_0 B_S}{\varepsilon_{\text{mach}}}\right)^{1/b} < \left(\frac{C_1}{\varepsilon_{\text{target}}}\right)^{1/a}}.$$

Numerical example. With $\beta_0 = 0.1$, $B_S = 1$, $\varepsilon_{\text{mach}} = 10^{-16}$, $b = 0.8$: - $t_{\text{freeze}} \approx (0.1/10^{-16})^{1.25} \approx 10^{18.75}$, astronomically large. Freezing is **not** a practical concern with double precision.

With $\beta_0 = 0.01$, $\varepsilon_{\text{mach}} = 10^{-4}$ (aggressive early stopping threshold), $b = 0.9$: - $t_{\text{freeze}} \approx (0.01/10^{-4})^{1.11} \approx 10^{2.22} \approx 166$, which is small enough to be concerning.

Status: PROVEN. The existence of a freezing time is guaranteed; whether it is premature depends on parameter choices.

1.4 Effect of Premature Freezing on Solution Quality

Theorem 12.3 (Suboptimality Gap Under Premature Freezing — PROVEN). If the gatekeeper freezes at $S_{t_{\text{freeze}}}$ and the student continues to converge toward $\theta^*(S_{t_{\text{freeze}}})$, the asymptotic Lyapunov gap is:

$$\Psi(S_{t_{\text{freeze}}}, \theta^*(S_{t_{\text{freeze}}})) - \Psi(S^*, \theta^*) \geq \lambda \cdot (\mathbb{E}_{V_0}[\ell(f_{\theta^*(S_{t_{\text{freeze}}})}, y)] - \mathbb{E}_{V_0}[\ell(f_{\theta^*}, y)]) \geq 0.$$

The gap is non-negative and equals zero only if $S_{t_{\text{freeze}}} = S^*$.

Proof. At the frozen gatekeeper $S_{t_{\text{freeze}}}$, the student converges to $\theta^*(S_{t_{\text{freeze}}})$, which minimizes the loss under the frozen distribution $P_{S_{t_{\text{freeze}}}}$. The gap on the reference distribution V_0 is generally positive because the frozen gatekeeper may exclude regions where the true fixed-point student θ^* performs differently. $\square$

1.5 Rate of Degradation

Corollary 12.1 (Degradation Rate Under Freezing). At time $t \geq t_{\text{freeze}}$, the convergence of Ψ_t stalls:

$$\Psi_t - \Psi^* = \Theta(1) \quad (\text{does not decay further}).$$

The improvement after freezing is:

$$\Delta\Psi_{\text{post-freeze}} = \Psi_{t_{\text{freeze}}} - \lim_{t \rightarrow \infty} \Psi_t = O(\alpha_{t_{\text{freeze}}}),$$

which comes only from the student continuing to train on the frozen distribution.

1.6 Mitigation Strategies

Strategy	Mechanism	Cost
Minimum learning rate	Clamp $\beta_t \geq \beta_{\min} > 0$	Prevents asymptotic convergence, introduces steady-state error $O(\beta_{\min})$
Cyclical learning rate	Periodically reset β_t to β_0	Requires tuning cycle period; may cause transient instability

Strategy	Mechanism	Cost
Plateau detection + boost	When $\ \Delta S_t\ < \delta$ for k consecutive steps, multiply β_t by $\gamma > 1$	Ad-hoc; may overshoot
Student-driven unfreezing	When student loss on V_0 stops improving, increase β_t temporarily	Couples unfreezing to observable metric
Polyak averaging on gatekeeper	Maintain $\bar{S}_t = \frac{1}{t} \sum_{i=1}^t S_i$	Reduces variance; may smooth past frozen states

2. Failure Mode 2: Backlog Problem (Memory Over-run)

2.1 Phenomenon

The memory bank M_t grows faster than the gatekeeper can score incoming samples. Unsourced samples accumulate in a backlog, delaying the gatekeeper's feedback and eventually stalling the self-evolution loop.

2.2 Formal Mechanism

Assumptions: - Samples arrive at rate r_{in} (samples per unit time). - The gatekeeper can score at rate r_{score} (samples per unit time). - The NEP student

retrains every K new samples.

Definition 2.1 (Backlog). The backlog at time t is:

$$B(t) = r_{\text{in}} \cdot t - r_{\text{score}} \cdot t = (r_{\text{in}} - r_{\text{score}}) \cdot t.$$

A backlog develops when $r_{\text{in}} > r_{\text{score}}$.

2.3 Gatekeeper Scoring Complexity

The gatekeeper’s per-sample scoring cost is:

$$T_{\text{score}}(x) = O(M) \quad (\text{evaluate all } M \text{ experts}) + O(K_S) \quad (\text{state lookup}) + O(d_\phi) \quad (\text{feature computation})$$

Total scoring throughput:

$$r_{\text{score}} = \frac{1}{T_{\text{score}}} = \Omega\left(\frac{1}{M + K_S + d_\phi}\right).$$

2.4 Critical Condition for Backlog

Theorem 12.4 (Backlog Instability — PROVEN). If:

$$r_{\text{in}} > \frac{C_{\text{compute}}}{M + K_S + d_\phi},$$

where C_{compute} is the available compute (FLOPs/second), the backlog grows without bound and the system becomes **unstable**: the memory bank con-

tains samples whose scores are based on a stale gatekeeper $S_{t-\tau}$ with delay $\tau = B(t)/r_{\text{score}} \rightarrow \infty$.

Proof. The scoring delay per sample is $\tau = B(t)/r_{\text{score}}$. When $r_{\text{in}} > r_{\text{score}}$, $B(t) \rightarrow \infty$ and $\tau \rightarrow \infty$. The gatekeeper used for sample x_i is S_{t_i} where $t_i = i - \tau_i$. As $\tau \rightarrow \infty$, the gatekeeper version used for scoring becomes arbitrarily old, violating the implicit assumption that S_t reflects current knowledge. $\square$

Status: PROVEN. This is a throughput calculation.

2.5 Effect on Self-Evolution Dynamics

Theorem 12.5 (Stale Gatekeeper Error — CONJECTURED). If sample x is scored by gatekeeper $S_{t-\tau}$ (stale by τ steps), the probability of misclassification is:

$$\mathbb{P}(\text{sign}(S_{t-\tau}(x) - \gamma) \neq \text{sign}(S_t(x) - \gamma)) \leq \frac{L_S \cdot \beta_{\max} \cdot \tau \cdot B_S}{\min(|S_t(x) - \gamma|, |S_{t-\tau}(x) - \gamma|)}.$$

For samples near the decision boundary ($|S_t - \gamma|$ small), the error can be $O(1)$ even for modest τ .

Proof sketch. By Lipschitz continuity (C3), $\|S_t - S_{t-\tau}\|_\infty \leq L_S \cdot \tau \cdot \beta_{\max} \cdot B_S$. The probability of sign change is bounded by the probability that the drift exceeds the margin. For small margin, this probability approaches 1. $\square$

Status: CONJECTURED. The Lipschitz bound is valid, but the probability estimate requires a model of the drift direction, which depends on the specific SCX update dynamics.

2.6 Backlog-Induced Distribution Shift

Proposition 12.1 (Backlog Distribution Bias — CONJECTURED).

When samples are scored with delay τ , the effective acceptance distribution is:

$$\tilde{P}_{S_t}(x, y) \propto \mathbf{1}\{S_{t-\tau}(x, y) \geq \gamma\} \cdot P_0(x, y).$$

This distribution is biased toward samples where $S_{t-\tau}$ overestimates reliability relative to S_t , because: - Samples scored by an older (less accurate) gatekeeper are more likely to be misclassified. - The direction of misclassification depends on whether S_t has increased or decreased relative to $S_{t-\tau}$.

If S_t is monotonically improving, then $S_t > S_{t-\tau}$ for most (x, y) (the gatekeeper becomes more generous over time). In this case, the stale gatekeeper is **more conservative**, rejecting some samples that S_t would accept. The effect is **reduced memory growth**, not incorrect acceptance.

If S_t oscillates, then $S_{t-\tau}$ may be higher than S_t in some regions, admitting samples that S_t would reject. This introduces **noisy samples** into M_t , degrading all downstream estimates.

2.7 Approximate Scoring for Throughput

To prevent backlog, the gatekeeper can use **approximate scoring**:

$$\tilde{S}_t(x, y) = S_t(\text{NN}_k(x), y),$$

where $\text{NN}_k(x)$ returns the k nearest neighbors of x among already-scored samples.

Theorem 12.6 (Approximation Error of NN Scoring — PROVEN).

Under C3 (Lipschitz gatekeeper) and assuming the already-scored samples form an ε -net of the input space:

$$\sup_{x,y} |\tilde{S}_t(x,y) - S_t(x,y)| \leq L_S \cdot \varepsilon.$$

Proof. By Lipschitz continuity, $|S_t(x) - S_t(\text{NN}(x))| \leq L_S \cdot \|x - \text{NN}(x)\| \leq L_S \cdot \varepsilon$ when the scored samples form an ε -net. $\square$

Status: PROVEN. This provides a principled approximation with controllable error.

2.8 Mitigation Strategies

Strategy	Effect	Trade-off
NN approximate scoring	$O(1)$ per sample (after index build)	$L_S \cdot \varepsilon$ score error
Batch scoring	Amortizes overhead	Increases delay for early samples in batch
Priority queue	Score high-uncertainty samples first	Requires uncertainty estimate before scoring (circular)
Parallel/distributed experts	Increases r_{score} linearly with workers	Communication overhead

Strategy	Effect	Trade-off
Gatekeeper distillation	Train a fast student gatekeeper $\tilde{S}$ to approximate S_t	Distillation error; requires periodic retraining
Adaptive sampling	Reduce r_{in} when backlog exceeds threshold	May miss novel samples

3. Failure Mode 3: Calibration Breakdown (Client Divergence)

3.1 Phenomenon

When two independent SCX clients (with different initial gatekeepers $S_0^{(1)}$, $S_0^{(2)}$ or different data streams) evolve their gatekeepers independently, their scoring functions may **diverge irreconcilably**: they assign opposite reliability judgments to the same samples, with no mechanism to reconcile.

3.2 Formal Setup

Definition 3.1 (Client Divergence). Two clients A and B evolve gatekeepers $S_t^{(A)}$ and $S_t^{(B)}$ from initial conditions $S_0^{(A)} \neq S_0^{(B)}$. Their **disagreement** at time t is:

$$D_t(A, B) = \mathbb{P}_{(x,y) \sim P_0}(\text{sign}(S_t^{(A)}(x, y) - \gamma) \neq \text{sign}(S_t^{(B)}(x, y) - \gamma)).$$

Divergence is **irreconcilable** if $\liminf_{t \rightarrow \infty} D_t(A, B) > 0$.

3.3 Existence of Multiple Fixed Points

Theorem 12.7 (Multiple Fixed Points — PROVEN). The SCX self-evolution dynamics can have multiple fixed points when: 1. The loss landscape is non-convex (multiple local minima for the student). 2. The gatekeeper’s acceptance policy can be self-consistent at different selectivity levels.

Proof by construction. Consider two fixed points: - **Fixed point A (conservative):** $S_A^*(x, y) \approx 0.3$ for all (x, y) . The gatekeeper is conservative, rejecting most samples. The memory bank is small, containing only the highest-confidence samples. The student is trained on this small, clean subset. **Self-consistent:** the conservative gatekeeper only admits samples it scores above 0.3, and the student trained on those samples predicts labels that the gatekeeper is comfortable with.

- **Fixed point B (permissive):** $S_B^*(x, y) \approx 0.7$ for all (x, y) . The gatekeeper is permissive, accepting most samples. The memory bank is large and diverse, including some noisy samples. The student is trained on this larger, noisier dataset. **Self-consistent:** the permissive gatekeeper admits most samples, and the student trained on those diverse samples makes predictions that maintain the permissive gatekeeper’s calibration.

Both are valid fixed points of the dynamics. They are **not** equivalent — they disagree on approximately 40% of decisions (samples with true reliability between 0.3 and 0.7). $\square$

Status: PROVEN. The existence of multiple fixed points follows from the non-convexity of the coupled system.

3.4 Basin of Attraction

Definition 3.2 (Basin of Attraction). The basin of attraction $\mathcal{B}(S^*, \theta^*)$ is the set of initial conditions (S_0, θ_0) from which the system converges to (S^*, θ^*) .

Conjecture 12.1 (Basin Structure). The state space partitions into basins of attraction:

$$\mathcal{Z} = \bigcup_{i=1}^{N_{\text{fp}}} \mathcal{B}_i \cup \mathcal{B}_{\text{boundary}},$$

where $\mathcal{B}_i$ are the basins of the N_{fp} fixed points, and $\mathcal{B}_{\text{boundary}}$ is a measure-zero set of initial conditions on the boundaries.

*Status: **CONJECTURED.** The basin structure for coupled SA systems is an active research area (Borkar, 2008, Chapter 9). The conjecture is plausible but unproven for the specific SCX dynamics.*

3.5 Condition for Irreconcilable Divergence

Theorem 12.8 (Irreconcilable Divergence Condition — PROVEN).

Two clients with initial conditions $(S_0^{(A)}, \theta_0^{(A)}) \in \mathcal{B}_i$ and $(S_0^{(B)}, \theta_0^{(B)}) \in \mathcal{B}_j$ with $i \neq j$ will converge to different fixed points with:

$$\lim_{t \rightarrow \infty} D_t(A, B) = D_{ij} > 0,$$

where D_{ij} is the disagreement rate between fixed points i and j .

Proof. By definition of basins of attraction, client A converges to fixed point i and client B to fixed point j . Since $i \neq j$, there exists at least one (x, y) where the fixed-point gatekeepers disagree. By continuity of the scoring function, the disagreement extends to a neighborhood of positive measure, so $D_{ij} > 0$. $\square$

Status: PROVEN (conditional on basin structure). The logic is sound; the only gap is the rigorous characterization of basin boundaries (Conjecture 12.1).

3.6 Calibration Degradation Rate

Proposition 12.2 (Calibration Drift Under Isolation — CONJECTURED). If two clients start from ε -close initial conditions but evolve on different data streams, their gatekeepers diverge at rate:

$$\mathbb{E}[D_t(A, B)] \leq \varepsilon \cdot \exp(\lambda_{\max} \cdot t),$$

where $\lambda_{\max}$ is the largest Lyapunov exponent of the deterministic part of the dynamics.

Proof sketch. Linearize the dynamics around the common trajectory. The difference $\delta_t = S_t^{(A)} - S_t^{(B)}$ evolves as $\delta_{t+1} \approx (I + \beta_t J_t) \delta_t + \text{noise}$, where J_t is the Jacobian of SCXUpdate. The largest eigenvalue of J_t determines the exponential divergence rate. $\square$

Status: CONJECTURED. The linearized analysis is standard for dynamical systems (Strogatz, 2018), but deriving $\lambda_{\max}$ for the specific SCX dynamics requires further work.

3.7 Mitigation Strategies

Strategy	Mechanism	Cost
Federated gatekeeper averaging	Periodically average $S_t^{(A)}$ and $S_t^{(B)}$	Requires communication; may pull both toward a worse compromise
Shared reference set	Both clients evaluate against the same M_0	Requires agreement on ground-truth labels for V_0
Consensus anchoring	Fix $\hat{C}(x)$ as a shared anchor; only evolve gatekeeper toward consensus, not away	Limits adaptation to client-specific data distributions
Cross-validation between clients	Each client's memory bank serves as validation for the other	Requires trust and data sharing
Divergence detection + reset	When $D_t(A, B) > \delta_{\text{threshold}}$, reset both to a common initialization	Loses learned information

4. Failure Mode 4: Adversarial Poisoning (Gatekeeper Corruption)

4.1 Phenomenon

An adversary injects carefully crafted samples into the data stream, causing the gatekeeper to systematically misclassify clean samples as noise (or vice versa). The self-evolution loop amplifies the corruption by training the student on poisoned data.

4.2 Threat Model

Definition 4.1 (Adversary Capabilities). The adversary can: - **Injection:** Add up to η_{adv} fraction of samples to the incoming data stream. - **Knowledge:** Know the current gatekeeper S_t , the expert models $\{f_m\}$, and the feature representation ϕ . - **Objective:** Maximize the false positive rate (clean $\rightarrow$ noise) or false negative rate (noise $\rightarrow$ clean) of the converged gatekeeper.

We assume the adversary does **not** control the expert models or the NEP training procedure — only the data fed to the gatekeeper.

4.3 Adversarial Sample Construction

Definition 4.2 (Consensus-Minimizing Adversarial Sample). To cause the gatekeeper to reject a clean sample (x, y_{true}) , the adversary constructs:

$$(x_{\text{adv}}, y_{\text{adv}}) = \arg \min_{(x', y') \in \mathcal{B}_\varepsilon(x)} \hat{C}(x') \quad \text{subject to} \quad \ell(f_m(x'), y') > \tau \text{ for most } m,$$

i.e., an ε -perturbation of a clean sample that maximally confuses the expert ensemble while remaining indistinguishable to the gatekeeper's feature representation.

4.4 Poisoning Amplification via Self-Evolution

The key danger in SCX self-evolution is **amplification**: one round of poisoning corrupts the gatekeeper, which admits more poisoned samples, which further corrupts the gatekeeper.

Theorem 12.9 (Poisoning Amplification Factor — CONJECTURED). If the adversary injects a fraction η_{adv} of poisoned samples at each round, the effective corruption after T rounds is:

$$\eta_{\text{eff}}(T) \geq \eta_{\text{adv}} \cdot \frac{1 - \gamma^T}{1 - \gamma},$$

where $\gamma = \mathbb{P}_{P_0}(S_t(x_{\text{adv}}, y_{\text{adv}}) \geq \gamma_t \mid \text{poisoned})$ is the probability that poisoned samples pass the gatekeeper.

When $\gamma > 0.5$ (poisoned samples are more likely to be accepted than rejected), the amplification diverges geometrically:

$$\eta_{\text{eff}}(T) \rightarrow 1 \quad \text{as} \quad T \rightarrow \infty.$$

Proof sketch. At each round, a fraction of poisoned samples enter M_t , corrupting the student. The corrupted student provides feedback that shifts

S_{t+1} toward accepting more poisoned samples (because the student agrees with the poisoned labels). This increases γ over time, creating a positive feedback loop. $\square$

Status: CONJECTURED. The geometric amplification depends on $\gamma > 0.5$, which in turn depends on the adversary’s ability to craft samples that both confuse experts and appear reliable to the current gatekeeper. This is a strong assumption that may not hold for well-regularized gatekeepers.

4.5 Detection Boundary for Adversarial Samples

Theorem 12.10 (Adversarial Detection Boundary — PARTIALLY PROVEN). Under the consensus score $\hat{C}(x)$ computed from M i.i.d. experts, an adversarial perturbation δ on a clean sample x changes the expected consensus by at most:

$$\mathbb{E}[|\hat{C}(x + \delta) - \hat{C}(x)|] \leq \frac{1}{M} \sum_{m=1}^M \mathbb{P}(\ell(f_m(x + \delta), y) > \tau \neq \ell(f_m(x), y) > \tau).$$

If each expert has a **robustness radius** ρ_m such that $\|\delta\| \leq \rho_m \implies f_m(x + \delta) = f_m(x)$, then for $\|\delta\| \leq \min_m \rho_m$, the consensus score is **unchanged** and the adversarial sample is undetectable by consensus alone.

Proof. The consensus changes only if the adversary’s perturbation flips at least one expert’s error indicator. If all experts are robust within radius $\rho_{\min}$, then no expert error indicator flips, and $\hat{C}$ is unchanged. $\square$

Status: PROVEN. This follows from the definition of $\hat{C}$ and provides a clean condition for undetectability.

4.6 Gatekeeper Robustness Certificate

Proposition 12.3 (Lipschitz-Based Robustness — PROVEN). Under C3 (Lipschitz gatekeeper with constant L_S), for any perturbation δ :

$$|S_t(x + \delta, y) - S_t(x, y)| \leq L_S \cdot \|\phi(x + \delta) - \phi(x)\| \leq L_S \cdot L_\phi \cdot \|\delta\|,$$

where L_ϕ is the Lipschitz constant of the feature map. The gatekeeper is **certifiably robust** for perturbations with $\|\delta\| \leq \frac{|S_t(x, y) - \gamma|}{L_S \cdot L_\phi}$.

Proof. Direct from C3 and the Lipschitz property of ϕ . $\square$

Status: PROVEN. This provides a certified radius within which adversarial perturbations cannot change the gatekeeper’s decision.

4.7 When the Student is the Vulnerability

Even if the gatekeeper is robust, the **NEP student** may be vulnerable. If poisoned samples enter M_t : 1. The student is trained on poisoned labels. 2. The student’s predictions on clean samples become corrupted. 3. The corrupted student feedback shifts the gatekeeper. 4. The cycle continues.

Theorem 12.11 (Student Poisoning Vulnerability — CONJECTURED). Under linear regression with ℓ_2 loss, if an η_{adv} fraction of training labels are flipped, the parameter error is:

$$\mathbb{E}[\|\hat{\theta} - \theta^*\|^2] \geq \eta_{\text{adv}}^2 \cdot \frac{\mathbb{E}[\|x\|^2]}{\lambda_{\min}(\Sigma)^2},$$

where $\Sigma = \mathbb{E}[xx^\top]$ is the feature covariance. For neural networks, the error may be larger due to the non-linear amplification of poisoned gradients.

Status: **CONJECTURED** (for neural networks). The linear case is standard (Hampel et al., 1986).

4.8 Mitigation Strategies

Strategy	Mechanism	Limitation
Robust aggregation (median-of-experts)	Replace mean consensus $\hat{C}$ with median or trimmed mean	Reduces effective M ; requires $> 50\%$ clean experts
Anomaly detection on $\phi(x)$	Flag samples with atypical features as potential adversarial	Adversary can constrain $\ \delta\ $ to stay in-distribution
Certified robustness (Lipschitz)	Bound $ S_t(x + \delta) - S_t(x) \leq L_S L_\phi \ \delta\ $	Loose bounds for large L_S, L_ϕ
Differential privacy in gatekeeper update	Add noise to ΔS_t to mask individual sample influence	Reduces convergence rate by $O(\sigma_{\text{DP}}^2)$
Student robust training	Train NEP with adversarial data augmentation	Increases training cost; may reduce clean accuracy
Outlier-resistant loss	Use Huber or Tukey loss for student training	Less statistically efficient than ℓ_2 for clean data
Human-in-the-loop for high-stakes decisions	Flag samples where $\hat{C}(x)$ and $S_t(x)$ disagree strongly	Does not scale

5. Interaction Effects: Compound Failures

5.1 Backlog + Premature Freezing

When backlog (Section 2) **and** premature freezing (Section 1) co-occur: - The backlog means the gatekeeper's decisions are stale ($S_{t-\tau}$ instead of S_t). - Premature freezing means the gatekeeper stops updating even as backlog grows. - **Compound effect:** The effective gatekeeper $S_{t-\tau}$ freezes at an even earlier (worse) state, and the delay τ grows without bound. The system becomes **doubly frozen** — neither the gatekeeper nor the scoring process can recover.

5.2 Adversarial Poisoning + Client Divergence

When adversarial poisoning (Section 4) targets one of two clients: - Client A (poisoned) converges to a corrupted fixed point. - Client B (clean) converges to a clean fixed point. - Their disagreement $D_t(A, B)$ grows over time. - **Compound effect:** The adversary can **maximize** $D_t(A, B)$ by injecting samples that are specifically chosen to push A's gatekeeper away from B's. This is a **targeted divergence attack**: the adversary exploits the multi-fixed-point structure.

5.3 Backlog + Adversarial Poisoning

When backlog means samples are scored by stale gatekeeper $S_{t-\tau}$: - The adversary can inject samples that are **benign** to the current gatekeeper S_t

but **malignant** to the stale gatekeeper $S_{t-\tau}$. - Since scoring uses $S_{t-\tau}$, the malignant samples pass through. - **Compound effect**: Backlog creates a **temporal attack surface** — the adversary can exploit the gap between the current and stale gatekeepers.

6. Diagnostic Tests for Each Failure Mode

6.1 Premature Freezing Detection

Test 1 (Gradient Norm Monitor). Track $\|\Delta S_t\|_{M_0}$ over a sliding window of W steps. If $\max_{i \in [t-W, t]} \|\Delta S_i\| < \varepsilon_{\text{diag}}$ for W consecutive windows, flag as frozen.

Test 2 (Student Improvement Correlation). Track the correlation between $\|\Delta S_t\|$ and $\Delta \Psi_{\text{student}, t}$. If the gatekeeper is frozen but the student is still improving, the correlation drops to zero.

6.2 Backlog Detection

Test 3 (Scoring Delay Monitor). Track $\tau(x) = t_{\text{current}} - t_{\text{scored}}$ for each sample. If $\text{median}(\tau) > \tau_{\text{threshold}}$, flag backlog.

Test 4 (Memory Growth vs. Scoring Rate). Track dN_t/dt (memory growth rate) and $d\text{Scored}_t/dt$ (scoring rate). If the former consistently exceeds the latter, backlog is growing.

6.3 Calibration Divergence Detection

Test 5 (Inter-Client Disagreement). Periodically compare gatekeeper decisions between clients on a shared validation set. If $D_t(A, B)$ is increasing and exceeds $\delta_{\text{threshold}}$, flag divergence.

Test 6 (Fixed-Point Distance Estimate). Estimate the distance between $S_t^{(A)}$ and $S_t^{(B)}$ in function space: $\|S_t^{(A)} - S_t^{(B)}\|_{M_0}$. A monotonically increasing trend suggests divergence to different fixed points.

6.4 Adversarial Poisoning Detection

Test 7 (Consensus-Gatekeeper Discrepancy). Track $\mathbb{E}[|\hat{C}(x) - S_t(x, y(x))|]$ on incoming samples. A sudden increase may indicate adversarial samples that confuse experts but are scored highly by the gatekeeper.

Test 8 (Expert Agreement Anomaly). Track the per-expert agreement matrix. Adversarial samples may cause unusual patterns (e.g., all experts agree on the wrong answer, or expert disagreement is bimodal rather than uniform).

Test 9 (Student Loss on Held-Out Clean Set). Monitor Ψ_{student} on V_0 . Poisoning should cause this to **increase** even as the training loss on M_t decreases (a classic sign of label poisoning).

7. Summary Table

Failure Mode	Cause	Effect	Critical Parameter	Severity	Provability
2. Backlog	$r_{\text{in}} > r_{\text{score}}$	Stale gate-keeper decisions; memory bank contaminated with misclassified samples	$r_{\text{in}}/r_{\text{score}}$	HIGH (practical for large M , high data rate)	Condition: PROVEN ; effect: CONJECTURED

Failure Mode	Cause	Effect	Critical Parameter	Severity	Provability
3. Client Divergence	Multiple fixed points; different initial conditions or data streams	Irreconcilable gate-keeper disagreement; no convergence to consensus	Delay (inter-fixed-point disagreement)	MODERATE (only in multi-client deployment)	Existence: PROVEN ; structure: CONJECTURED
4. Adversarial Poisoning	Adversary injects crafted samples	Gatekeeper corruption amplifies through self-evolution loop	$\gamma_{\text{adv}}, \gamma$ (acceptance rate of poisoned samples)	HIGH (severe with amplification)	Amplification: CONJECTURED ; robustness certificate: PROVEN

7.1 Risk Matrix

		Likelihood		
		Low	Med	High
Severity	High		2	4

Moderate		1		3			
Low							

- **High risk, high likelihood:** Backlog (2), Adversarial poisoning (4)
- **High risk, moderate likelihood:** —
- **Moderate risk, low likelihood:** Premature freezing (1)
- **Moderate risk, moderate likelihood:** Client divergence (3)

7.2 Design Recommendations

1. **Prevent premature freezing:** Always maintain $\beta_{\min} > 0$ or use cyclical learning rates. Monitor $\|\Delta S_t\|$ as a diagnostic.
2. **Prevent backlog:** Implement NN approximate scoring with bounded error. Monitor scoring delay. Trigger adaptive sampling when backlog exceeds threshold.
3. **Prevent client divergence:** Use a shared reference set M_0 with agreed-upon labels for V_0 . Periodically synchronize gatekeepers in federated deployments.
4. **Prevent adversarial poisoning:** Use Lipschitz-regularized gatekeepers with certified robustness radii. Train the NEP student with robust loss functions. Monitor consensus-gatekeeper discrepancy on held-out data.

End of 12_edge_cases.md — Edge Cases and Failure Modes

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Changelog

Date	Defect	Change	Severity
2026-06-28	B4	Renamed gatekeeper update rate from to : The symbol η is reserved throughout SCX theory for the global noise rate $\eta = P(Z = 1)$. All local uses of $\eta(t)$ as gatekeeper update rate replaced with β_t . Notation note added at document header and Theorem 12.1. Fixes FAIL-2 from cross-reference audit.	FATAL